

Complexity & Algorithms, Spring '26

Matchings
Class recap



What's left?

- For me:
 - grading essay + final HW
 - finishing last few regrades
 - writing a final!
- For you:
 - Oral HW: by Thursday
 - Study: practice exam + extra problems posted

Final topics

First half:

- Recursion
- DP
- Greed + approximation
- NP-Hardness + reductions

Second half!

- Graph algorithms:
 - BFS / DFS
 - Top sort (+DP)
 - Shortest Paths
- LPs
- Lower bounds
- FFTs
- Max Flow

Final exam format:

- 7 questions
- I'll take 5 highest scores
- Cheat sheets - hand written -
are allowed

Last topic: Flows + matchings
We saw max flow used to
find matchings in bipartite
graphs:

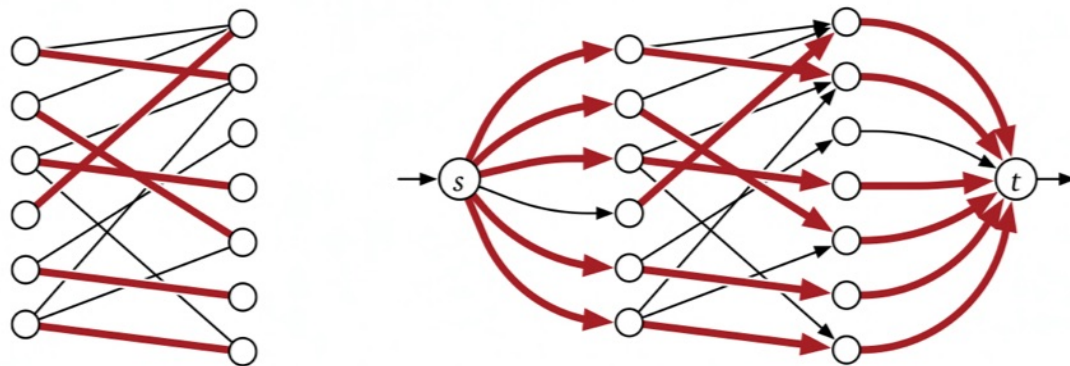
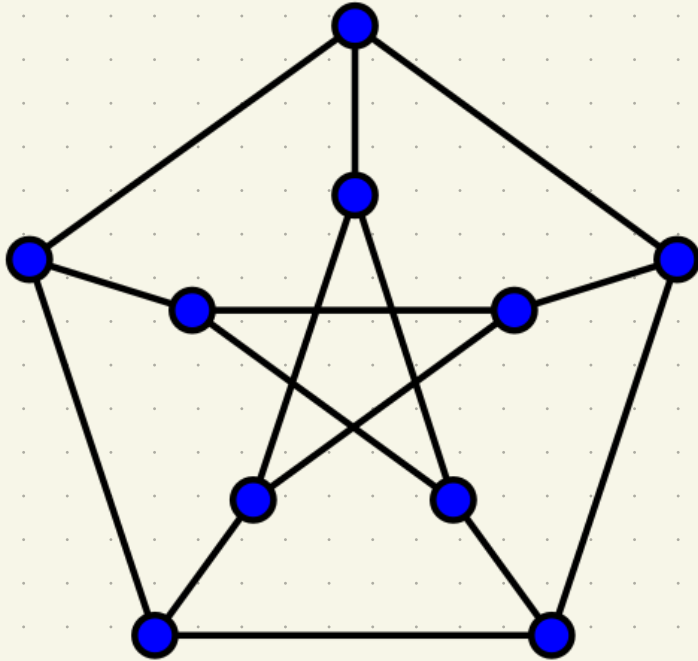


Figure 11.2. A maximum matching in a bipartite graph G , and the corresponding maximum flow in G' .

Why does this fail in general graphs?

First:
maximum

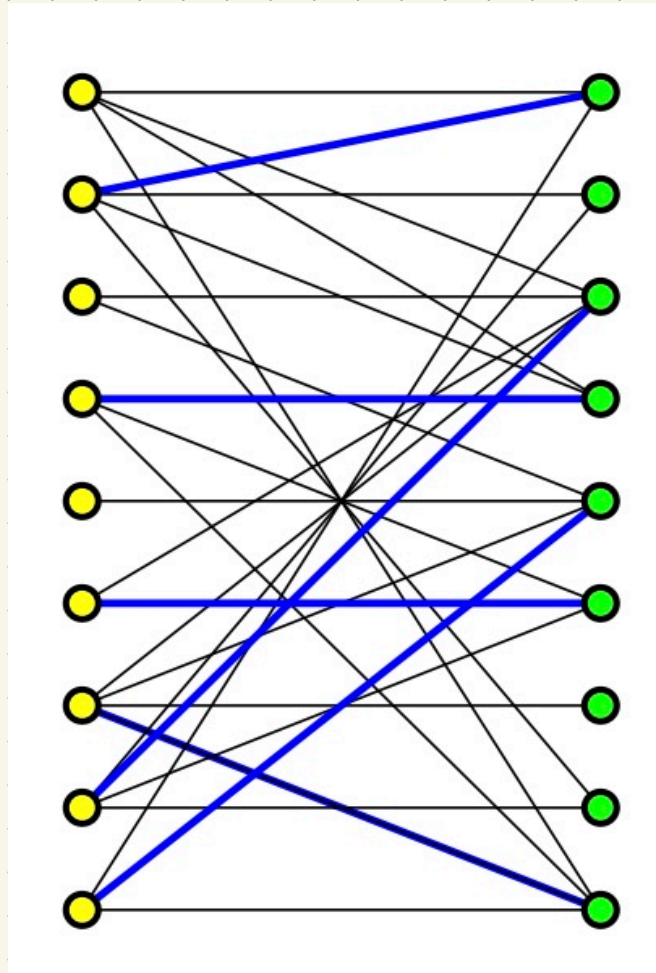


vs

maximal

In bipartite graphs, residual networks
find augmenting paths!

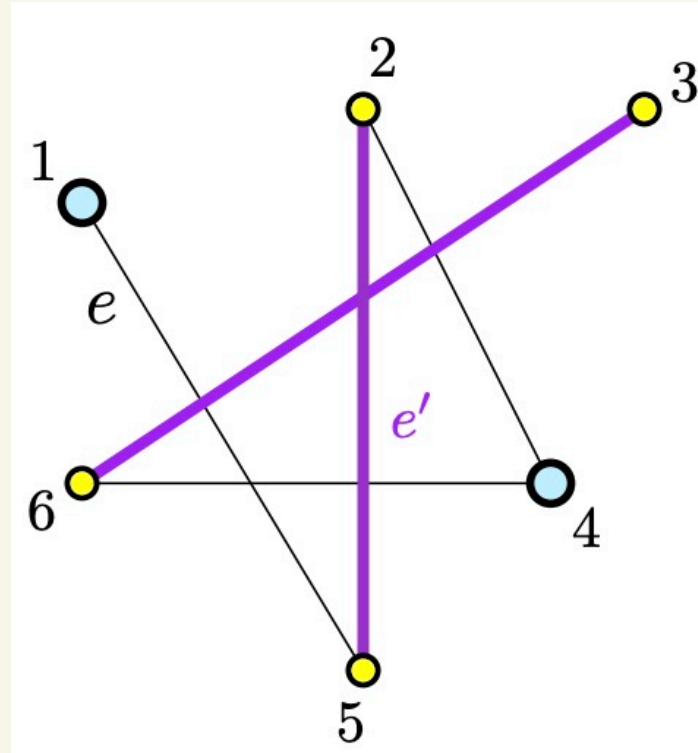
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Similar techniques for general graphs
(but no flows!)

Find an
"alternating path":

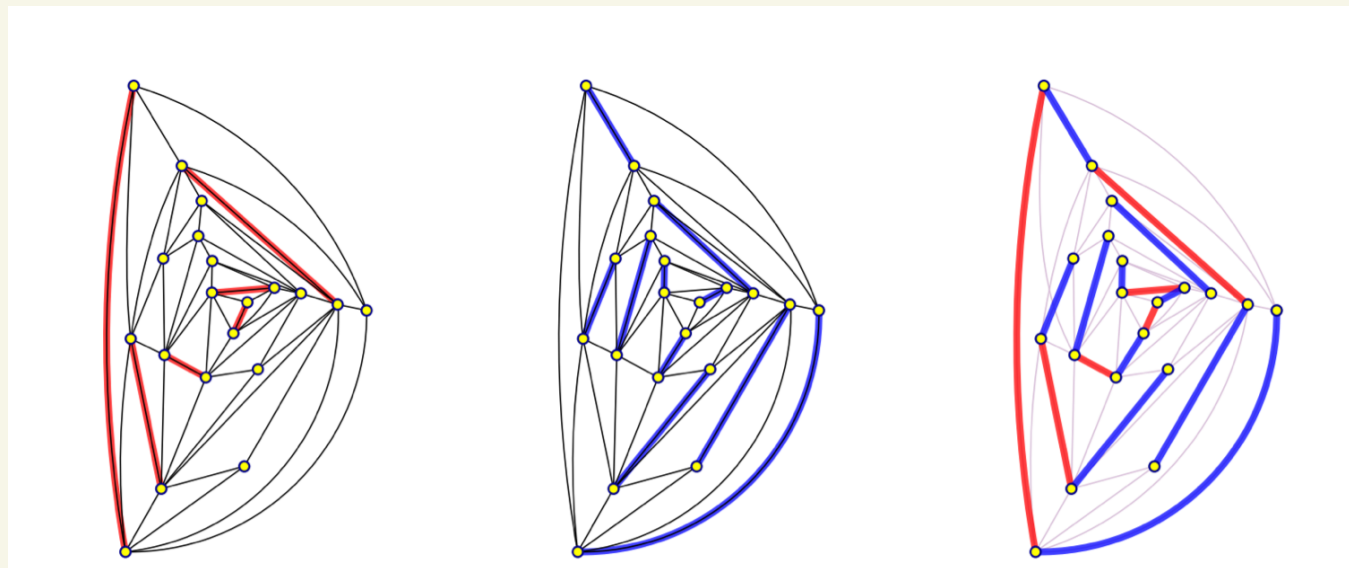


Berge's Optimality Criteria:

A vertex is free if no edge from the matching is incident to it.

A matching M is optimum $\Leftrightarrow$ no M -augmenting paths.

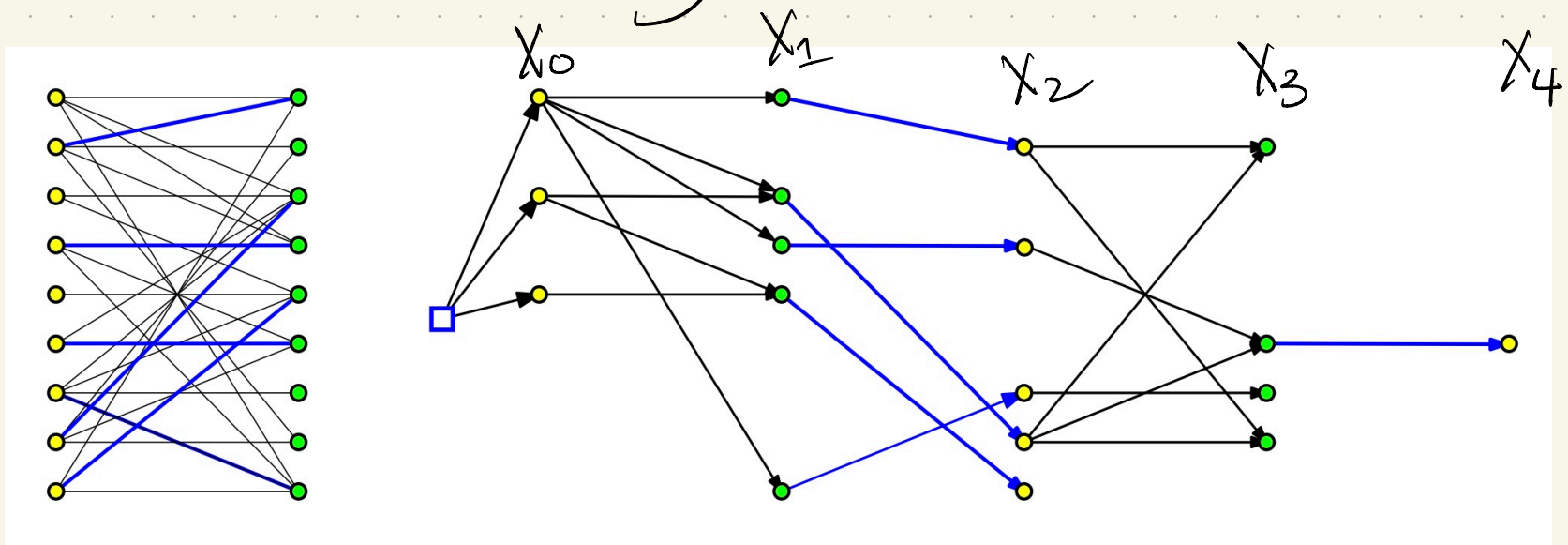
Why?



Algorithm using modified BFS:

Take a matching, & start a BFS
from open vertices in L :

Then alternate matching &
non-matching edges in levels.



Goal: Find path from open $v \in X_0$ to
open $w \in X_{2i+1}$.

Take away:

An algorithm - runtime?

Rest of chapter: faster ways: $O(E^{1+o(1)})$

General graphs: $O(EV^2)$